



CHAPTER 9-IMPLEMENTING SECURITY



Table of Contents

Exercise 1 – Implementing DLP	2
Exercise 2 – Implementing Anti-Malware Policies.....	7

Exercise 1 – Implementing DLP

Objective:

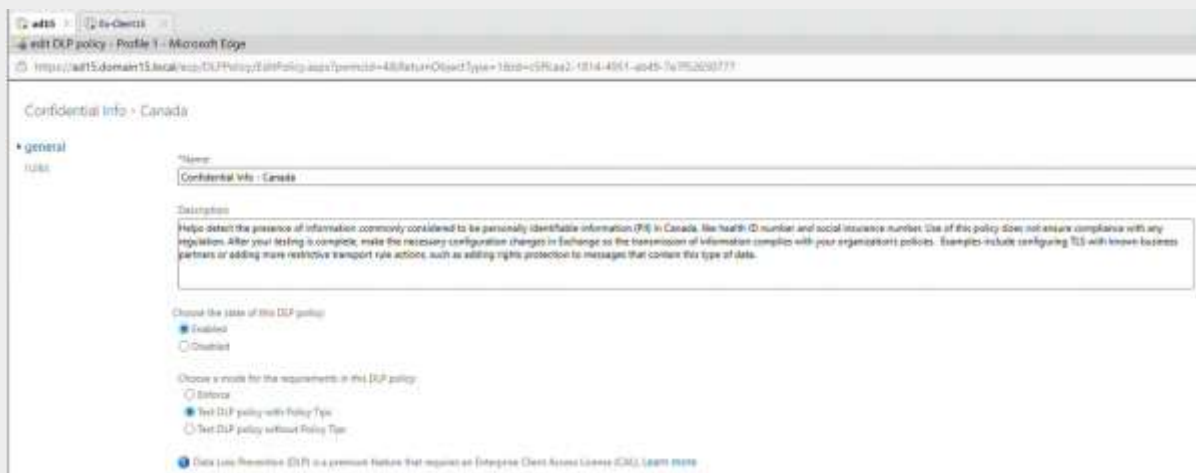
Practice creating, customizing, and reviewing DLP policies using both the Exchange Admin Center and PowerShell.

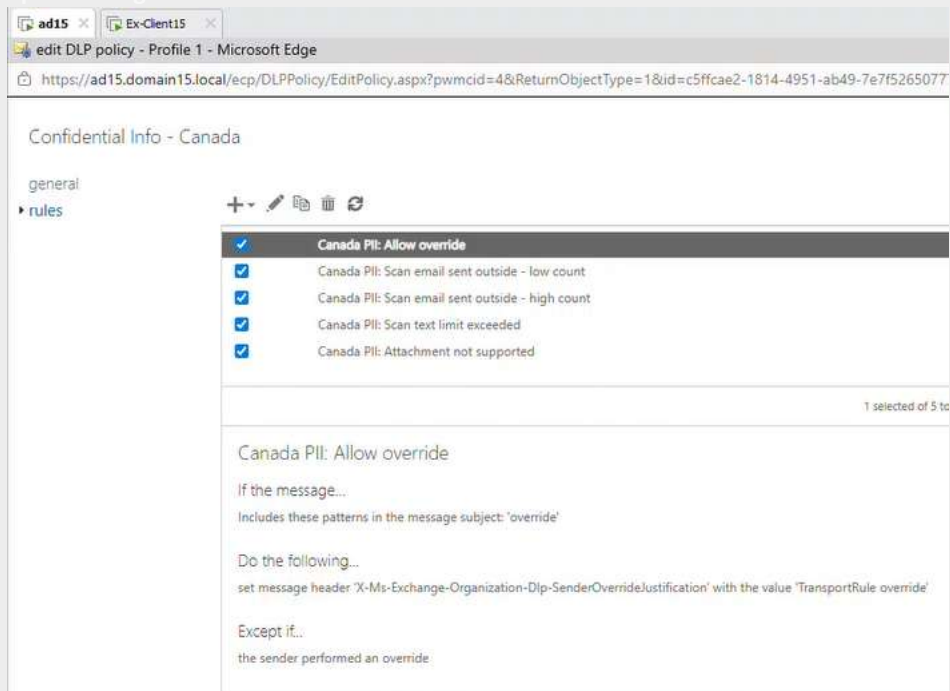
Tasks:

Using the Exchange Admin Center

1. Create a new **DLP policy** based on the following:

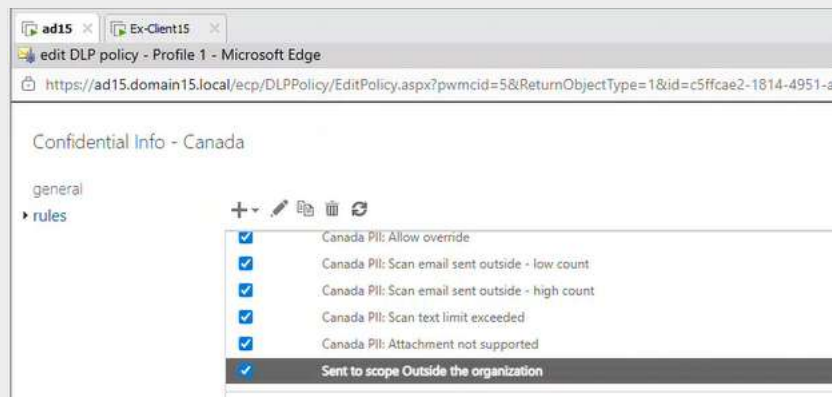
- Name: Confidential Info - Canada
- Template: Canada Personally Identifiable Information (PII)
- Enforcement Mode: **Test with Policy Tips**

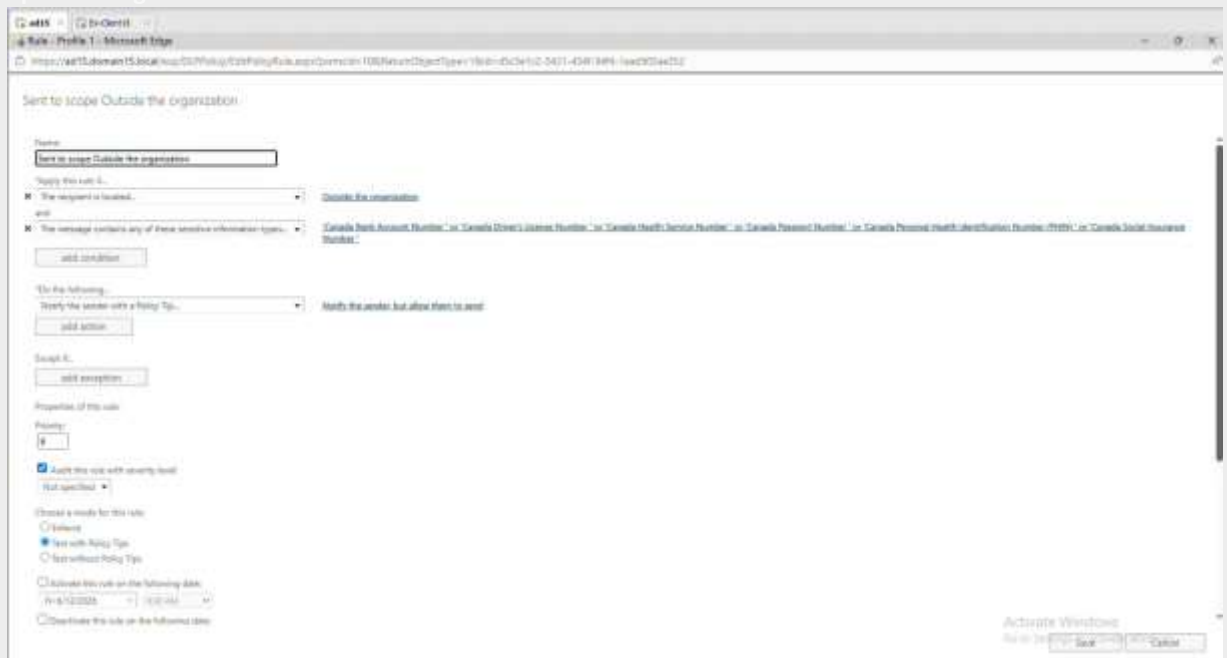




2. Modify the policy to:

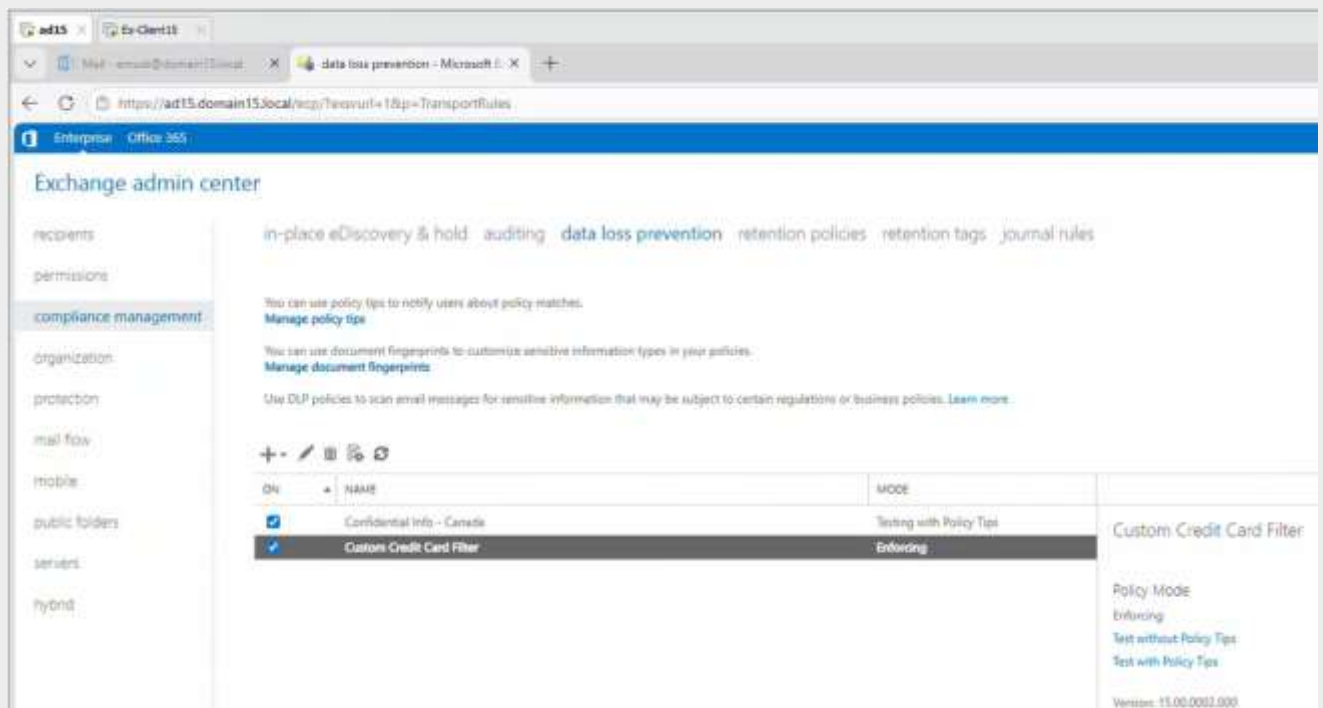
- Add a new custom rule that **warns** the sender if the message includes sensitive information.
- Apply the rule to emails sent **outside** the organization.



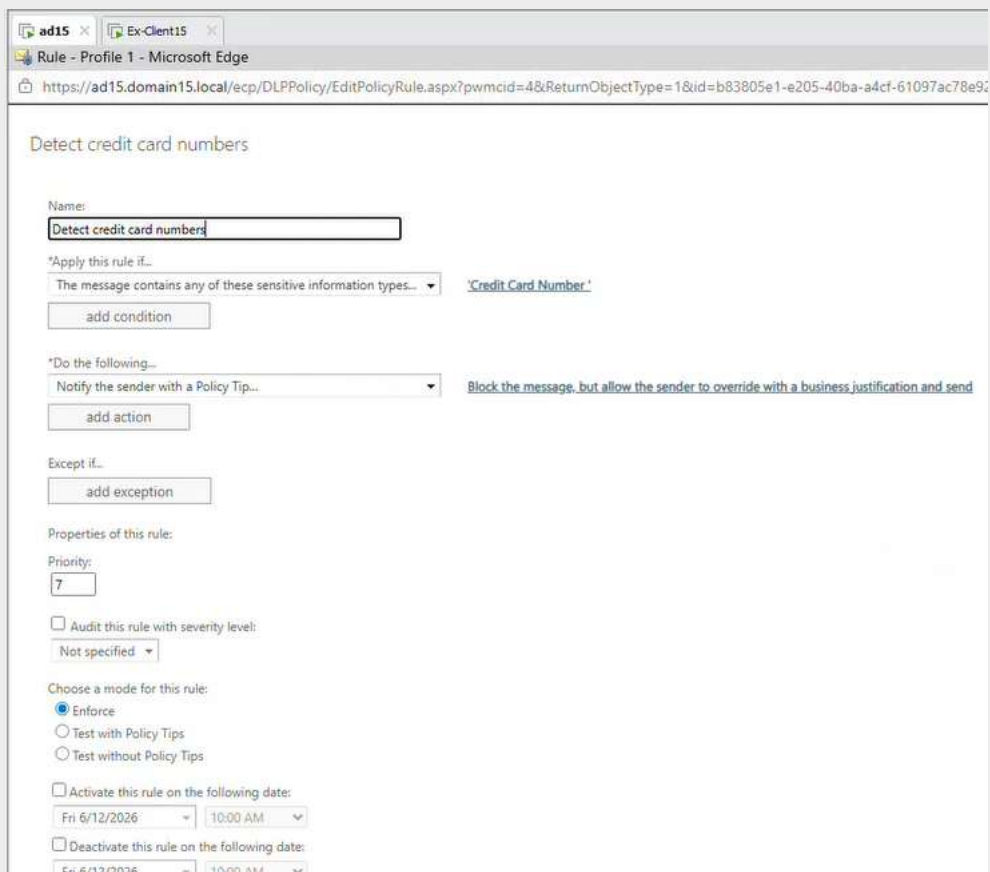
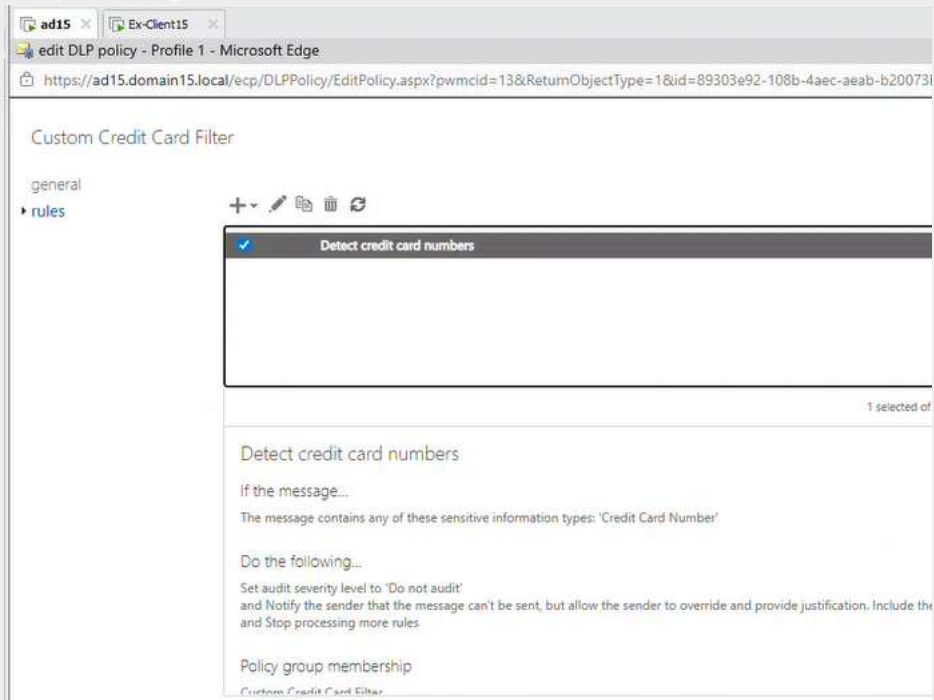


3. Create another DLP policy from scratch (**without using a template**):

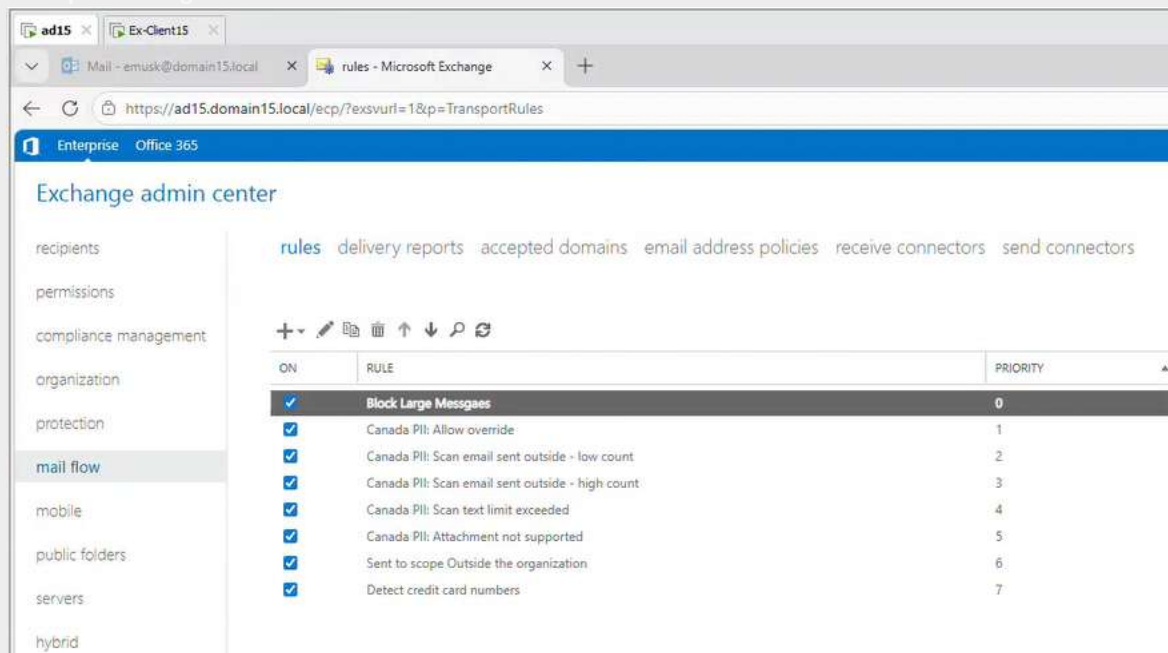
- Name: Custom Credit Card Filter
- Add a rule that:
 - Detects credit card numbers
 - Blocks the message if found
 - Allow the sender to override with justification



Lab 9 – Implementing DLP & Anti-Malware Policies

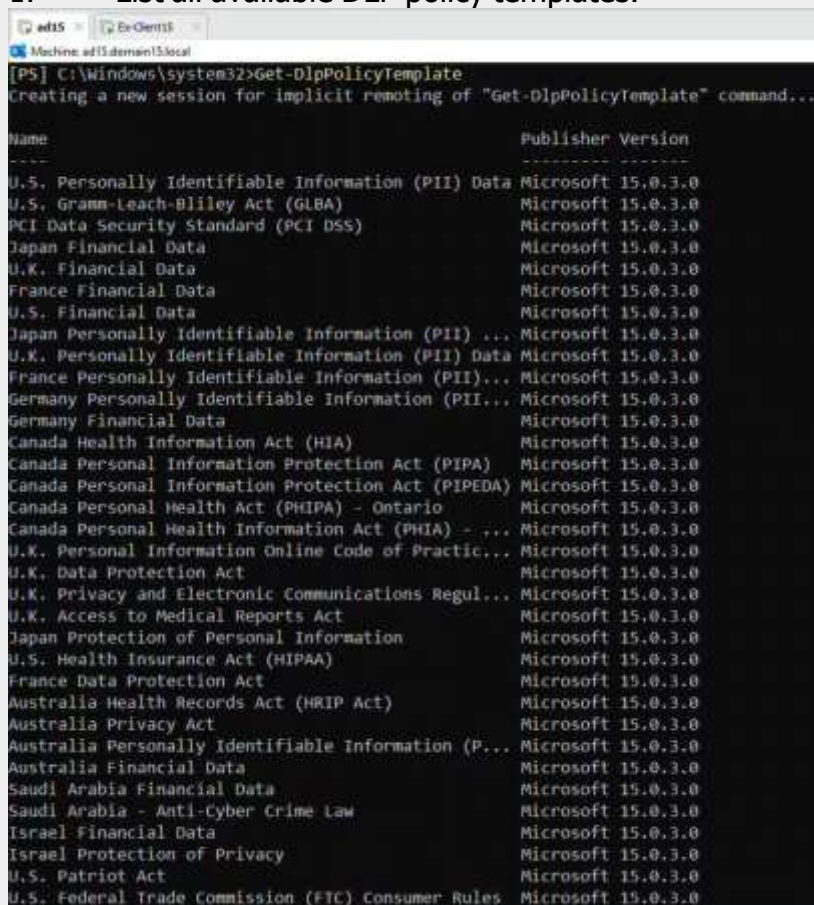


4. Review all DLP policies and analyze the rules created



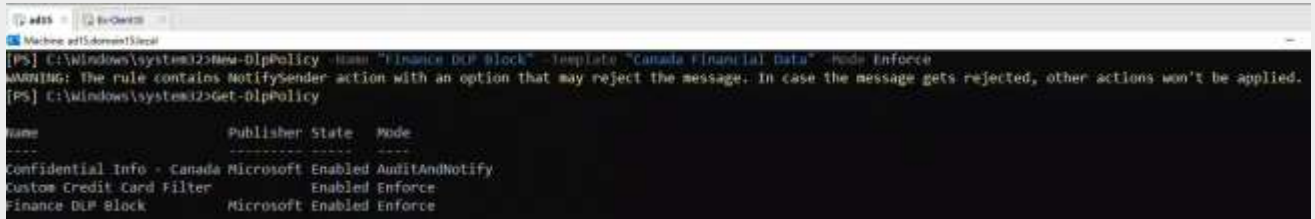
Using the PowerShell Exchange Management Shell (EMS)

1. List all available DLP policy templates.



2. Create a DLP policy:

- Name: Finance DLP Block
- Template: Canada Financial Data
- Enforcement Mode: Enforce

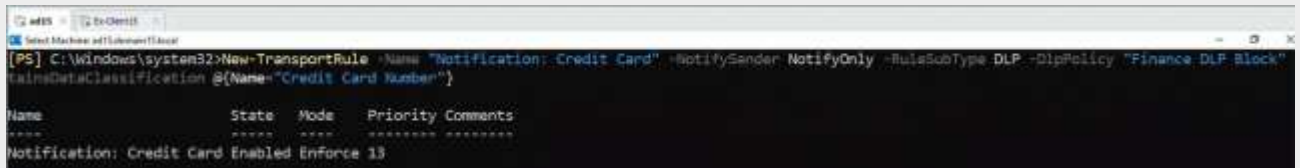


```
[PS] C:\Windows\system32>New-DlpPolicy -Name "Finance DLP Block" -Template "Canada Financial Data" -Mode Enforce
WARNING: The rule contains NotifySender action with an option that may reject the message. In case the message gets rejected, other actions won't be applied.
[PS] C:\Windows\system32>Get-DlpPolicy
```

Name	Publisher	State	Mode
Confidential Info - Canada	Microsoft	Enabled	AuditAndNotify
Custom Credit Card Filter		Enabled	Enforce
Finance DLP Block	Microsoft	Enabled	Enforce

3. Create a transport rule linked to the Finance DLP Block policy that:

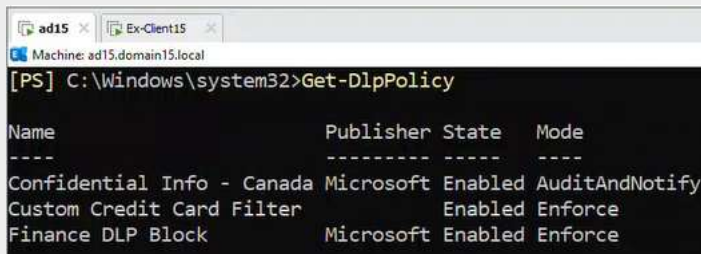
- Notify the sender when the content includes a credit card number.
- Applies only to external recipients.



```
[PS] C:\Windows\system32>New-TransportRule -Name "Notification: Credit Card" -NotifySender NotifyOnly -RuleSubType DLP -DlpPolicy "Finance DLP Block" -TrainsDataClassification @(Name="Credit Card Number")
```

Name	State	Mode	Priority	Comments
Notification: Credit Card	Enabled	Enforce	13	

4. List all DLP policies currently deployed in the organization.



```
[PS] C:\Windows\system32>Get-DlpPolicy
```

Name	Publisher	State	Mode
Confidential Info - Canada	Microsoft	Enabled	AuditAndNotify
Custom Credit Card Filter		Enabled	Enforce
Finance DLP Block	Microsoft	Enabled	Enforce

Exercise 2 – Implementing Anti-Malware Policies

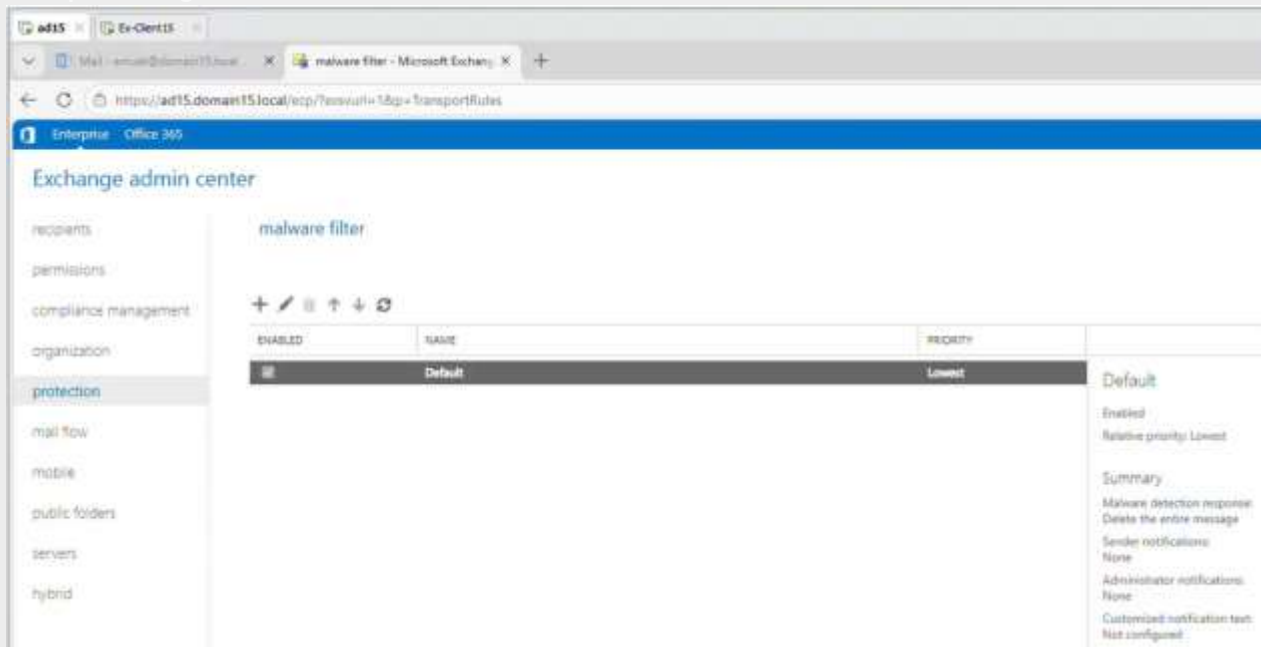
Objective:

Explore the creation and customization of **anti-malware policies using EAC and PowerShell.**

Tasks:

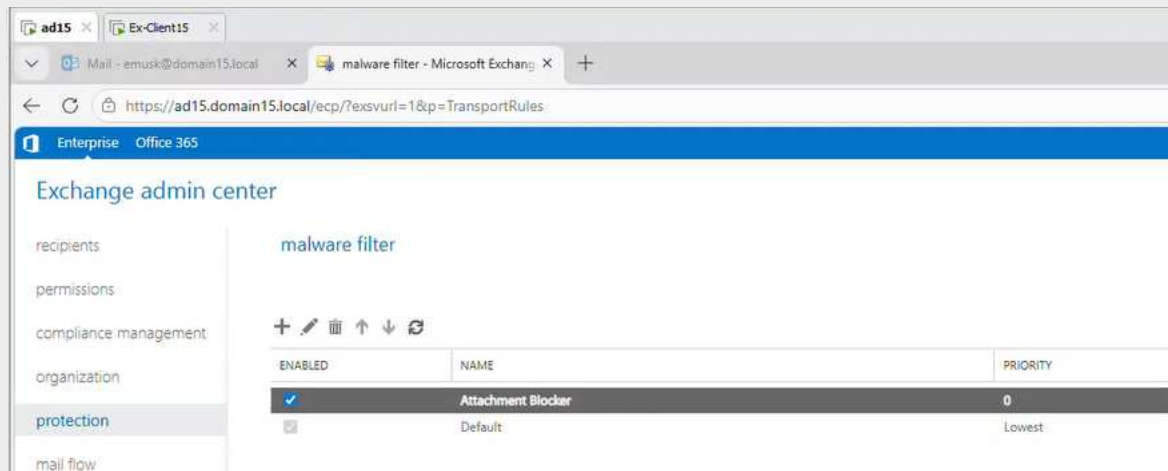
Using the Exchange Admin Center:

1. Review the default anti-malware policy under Protection > Malware Filter.

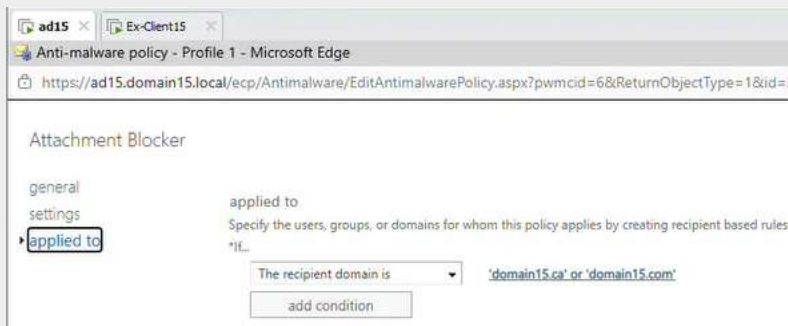
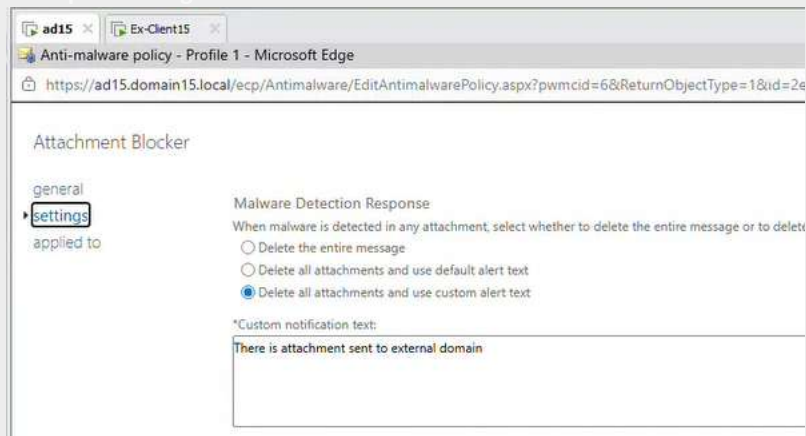


2. Create a new policy with:

- Name: **Attachment Blocker**
- Action: Delete the message and notify the sender
- Condition: Applies to messages sent to an external domain (e.g., @domainXX.com)

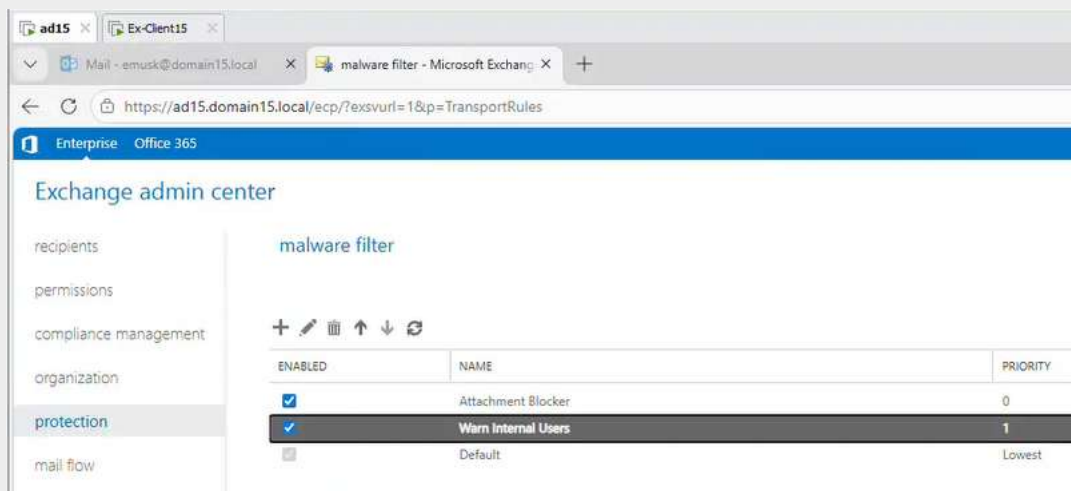


Lab 9 – Implementing DLP & Anti-Malware Policies



3. Create another policy named **Warn Internal Users** that:

- Deletes only the infected attachment.
- Send a notification to the administrator.



Lab 9 – Implementing DLP & Anti-Malware Policies

ad15 x Ex-Client15 x

Anti-malware policy - Profile 1 - Microsoft Edge

https://ad15.domain15.local/ecp/Antimalware/EditAntimalwarePolicy.aspx?pwmcid=11&ReturnObjectType=1&id

Warn Internal Users

general

settings

applied to:

Malware Detection Response

When malware is detected in any attachment, select whether to delete the entire message or to delete the attachments.

☐ Delete the entire message

☐ Delete all attachments and use default alert text

☒ Delete all attachments and use custom alert text

*Custom notification text:

The attachment is infected

Notifications

Sender Notifications

Sends a message to the sender of the undelivered message.

☐ Notify internal senders

☐ Notify external senders

Administrator Notifications

Sends a message to the administrator of the undelivered message.

☒ Notify administrator about undelivered messages from internal senders

Administrator email address:

administrator@domain15.local

☐ Notify administrator about undelivered messages from external senders

Administrator email address:

ad15 x Ex-Client15 x

Anti-malware policy - Profile 1 - Microsoft Edge

https://ad15.domain15.local/ecp/Antimalware/EditAntimalwarePolicy.aspx?pwmcid=11&ReturnObjectType=1&id

Warn Internal Users

general

settings

applied to

Specify the users, groups, or domains for whom this policy applies by creating recipients.

*if...

The recipient domain is domain15.local

add condition

Except if...

add exception

4. Verify all created anti-malware policies and their associated settings.

ad15 x Ex-Client15 x

Mail - emusk@domain15.local x malware filter - Microsoft Exchange x +

https://ad15.domain15.local/ecp/?exsvurl=1&up=TransportRules

Enterprise Office 365

Exchange admin center

recipients

permissions

compliance management

organization

protection

mail flow

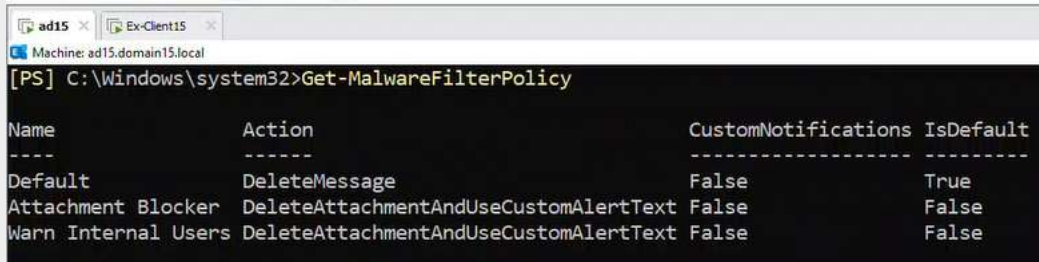
malware filter

+ - - - - -

ENABLED	NAME
☒	Attachment Blocker
☒	Warn Internal Users
☐	Default

Using the Exchange Management Shell (EMS):

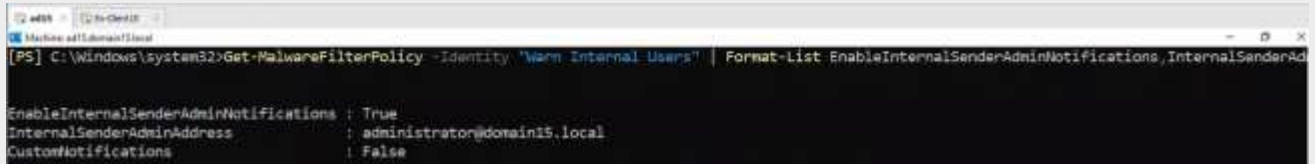
1. List all existing anti-malware policies.



```
[PS] C:\Windows\system32>Get-MalwareFilterPolicy
```

Name	Action	CustomNotifications	IsDefault
Default	DeleteMessage	False	True
Attachment Blocker	DeleteAttachmentAndUseCustomAlertText	False	False
Warn Internal Users	DeleteAttachmentAndUseCustomAlertText	False	False

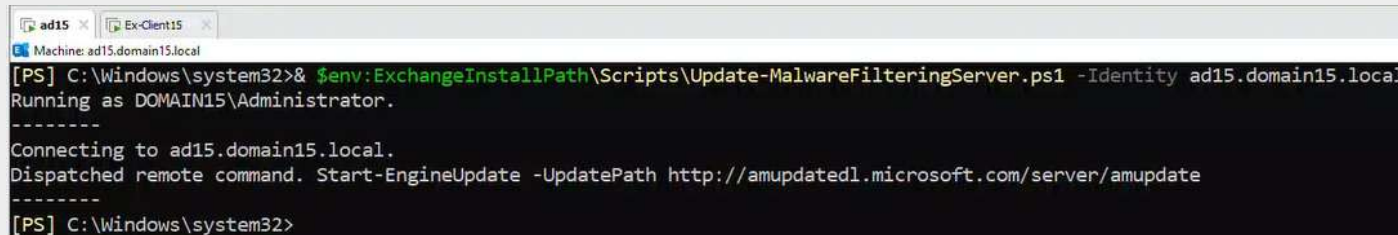
2. Review the notification settings applied to a selected anti-malware policy.



```
[PS] C:\Windows\system32>Get-MalwareFilterPolicy -Identity 'Warn Internal Users' | Format-List EnableInternalSenderAdminNotifications, InternalSenderAdminAddress, CustomNotifications
```

```
EnableInternalSenderAdminNotifications : True
InternalSenderAdminAddress             : administrator@domain15.local
CustomNotifications                    : False
```

3. Force a malware definition update using the appropriate script.



```
[PS] C:\Windows\system32>& $env:ExchangeInstallPath\Scripts\Update-MalwareFilteringServer.ps1 -Identity ad15.domain15.local
```

```
Running as DOMAIN15\Administrator.
```

```
-----
```

```
Connecting to ad15.domain15.local.
```

```
Dispatched remote command. Start-EngineUpdate -UpdatePath http://amupdated1.microsoft.com/server/amupdate
```

```
-----
```

```
[PS] C:\Windows\system32>
```